

Daily Problem: 07–Feb–2026

Problem Statement

Fix $k \in \mathbb{N}$ with $k > 0$. Consider the language over $\Sigma = \{0, 1\}$ consisting of all binary encodings of numbers divisible by k (read most-significant bit first). Formally, let $\text{val}(w)$ be the integer represented by a binary string w , and define

$$L_k = \{ w \in \{0, 1\}^* \mid \text{val}(w) \equiv 0 \pmod{k} \}.$$

Show that any DFA recognizing L_k must have at least k states.

Proof

Let $D = (Q, \{0, 1\}, \delta, q_0, F)$ be a DFA recognizing L_k . We prove $|Q| \geq k$ by showing that D must be able to distinguish k different remainders modulo k .

Step 1: A key arithmetic identity

If a binary string w represents the number $\text{val}(w)$, then appending a bit $b \in \{0, 1\}$ produces the number

$$\text{val}(wb) = 2 \text{val}(w) + b.$$

More generally, for any $t \geq 0$,

$$\text{val}(w 0^t) = 2^t \text{val}(w).$$

Step 2: Distinguishing prefixes via Myhill–Nerode

For each remainder $r \in \{0, 1, \dots, k-1\}$, choose some binary string x_r such that

$$\text{val}(x_r) \equiv r \pmod{k}.$$

(For example, one can take x_r to be the usual binary representation of r .)

We claim that the k prefixes $x_0, x_1, \dots, x_{k-1}$ must lead the DFA to k distinct states. Equivalently, we claim that for $r \neq s$,

$$\hat{\delta}(q_0, x_r) \neq \hat{\delta}(q_0, x_s).$$

Assume for contradiction that for some $r \neq s$,

$$\hat{\delta}(q_0, x_r) = \hat{\delta}(q_0, x_s).$$

Then for every suffix $y \in \{0, 1\}^*$, the DFA would behave identically after reading $x_r y$ and $x_s y$; in particular,

$$x_r y \in L_k \iff x_s y \in L_k.$$

We will produce a specific suffix y that separates them.

Step 3: Constructing a distinguishing suffix

Since $r \neq s$ modulo k , we have $r - s \not\equiv 0 \pmod{k}$. Pick an integer $t \geq 0$ such that 2^t is invertible modulo k on the difference $r - s$. A clean way to ensure this is to choose t so that

$$\gcd(2^t, k) = 1,$$

which holds for example when k is odd (take $t = 0$), and in general one can take t large enough so that all factors of 2 are absorbed and work modulo the odd part of k . An even simpler argument avoids this subtlety entirely by using a different suffix construction below.

Suffix that always works. Instead of powers of two, we use the following standard trick: choose a number m such that

$$m \equiv -r \pmod{k}.$$

Let y be the binary representation of m . Then

$$\text{val}(x_r y) \equiv r + m \equiv 0 \pmod{k},$$

so $x_r y \in L_k$.

On the other hand,

$$\text{val}(x_s y) \equiv s + m \equiv s - r \not\equiv 0 \pmod{k},$$

since $r \neq s$. Therefore $x_s y \notin L_k$.

Thus the suffix y distinguishes x_r and x_s , contradicting the assumption that the DFA reached the same state after reading x_r and x_s .

Step 4: Concluding the lower bound

We have shown that the k strings $x_0, \dots, x_{k-1}$ must reach k distinct states in any DFA for L_k . Therefore,

$$|Q| \geq k.$$

Conclusion

Any deterministic finite automaton recognizing the set of binary encodings of integers divisible by k must have at least k states.